Chapter 1 — Periodic Table and Periodic Properties

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Across a period, atomic radius: (a) Increases (b) Zero (c) Decreases (d) Same

2. Down a group, ionization energy: (a) Decreases (b) Zero (c) Increases (d) Same

3. Most electronegative element: (a) N (b) F (c) O (d) Cl

4. Na + cold water gives: (a) $NaOH + H_2$ (b) $NaH + O_2$ (c) $NaCl + H_2$ (d) $Na_2O + H_2$

5. Mg + steam gives: (a) $Mg(OH)_2 + O_2$ (b) $MgCl_2$ (c) MgH_2 (d) $MgO + H_2$

6. Al_2O_3 is: (a) Neutral (b) Acidic (c) Amphoteric (d) Basic

7. SO_3 is: (a) Neutral (b) Acidic (c) Basic (d) Amphoteric

8. $SiCl_4$ in water is: (a) Basic (b) Acidic (c) Insoluble (d) Neutral

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is a cation smaller than its parent atom?

2. Why does nitrogen have a higher ionization energy than oxygen?

3. Why is chlorine’s electron affinity more negative than fluorine’s?

4. Why does sodium react more vigorously with water than magnesium?

5. Why is aluminium oxide amphoteric?

6. Why does $SiCl_4$ hydrolyse while NaCl does not?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the trends in atomic and ionic radii across a period and down a group. **(4)**

(b) Explain periodic trends in ionization energy, electron affinity and electronegativity. **(4)**

Q.4 (a) Compare the reactions of sodium and magnesium with water, oxygen and chlorine. **(4)**

(b) Classify Period 3 oxides and chlorides and explain the variation in oxidation number. **(4)**

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